

Some Representation Theory

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1 Maschke's Theorem

In this section, we present the proof of Maschke's Theorem in two different ways, both using an averaging trick. Recall that Maschke's theorem states that every finite-dimensional representation of a finite group is completely reducible, i.e., it can be written as a direct sum of irreducible subrepresentations. In the first approach, we show that any finite-dimensional representation is equivalent to a unitary representation, and therefore every finite-dimensional representation is either irreducible or decomposable, from which the theorem follows. In the second approach, we first show directly, using the averaging trick, that every subrepresentation has a complementary subrepresentation. We now proceed to the statement and proof of the theorem.

Theorem 1. *Every representation of a finite group is completely reducible.*

1.1 Proof -1

Lemma 1. *Let $\phi : G \rightarrow \text{GL}(V)$ be a unitary representation of a group G where V is a finite dimensional $\mathbb{C}$ - vector space. Then ϕ is either irreducible or decomposable.*

Proof. If ϕ is not irreducible, then there is a non-zero proper G - invariant subspace W of V . Then show that $W^\perp$ is a G - invariant subspace of V so that $V = W \oplus W^\perp$ and V is decomposable. $\square$

Lemma 2. *Every representation of a finite group G is equivalent to a unitary representation.*

Proof. Let $\phi : G \rightarrow \text{GL}(V)$ be a finite dimensional representation and $T : V \rightarrow \mathbb{C}^n$ be a vector space isomorphism. Then we get a representation ρ of G on $\mathbb{C}^n$ by setting $\rho_g = T\phi_g T^{-1}$. This representation is equivalent to ϕ . Let $\langle \cdot | \cdot \rangle$ be the standard inner product on $\mathbb{C}^n$. Then the map from $V \times V$ to $\mathbb{C}$ defined by

$$(v, w) = \sum_{g \in G} \langle \rho_g v | \rho_g w \rangle$$

is an inner product and the representation ρ is unitary with respect to this inner product. $\square$

From the above two lemmas it must be clear that every representation of a finite group is either irreducible or decomposable. (Equivalent representations preserve the properties such as irreducibility, decomposable, etc.). But this is false for infinite groups. For example, the representation $\phi : \mathbb{Z} \rightarrow \text{GL}_2(\mathbb{C})$ given by

$$\phi(n) = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$$

This representation is not irreducible and indecomposable (Verify).

Now we come to the proof of Maschke's theorem.

Proof. The proof proceeds by induction on the dimension of a G -module. Let $\phi : G \rightarrow \text{GL}(V)$ be a representation of a finite group G . If $\dim V=1$, it is obvious. Otherwise by the above lemma, it is either irreducible or decomposable. If it is irreducible, we are done. If not, it is decomposable, say $V = U \oplus W$. Now by induction hypothesis U and W are completely reducible and the theorem follows. $\square$

1.2 Proof-2

In this approach we first prove the following theorem. All the notations are same as above.

Theorem 2. *If U is a submodule of the G -module V , then V is decomposable, i.e., there exists a G -submodule W such that $V = U \oplus W$.*

Proof. First we choose a subspace W_0 of V such that $V = U \oplus W_0$ as a vector space. Let ϕ denote the projection of V onto U . We like to modify the projection such to construct a G -module homomorphism with image U . For that, we define $T : V \rightarrow V$ by

$$Tv = \frac{1}{|G|} \sum_{g \in G} g^{-1} \phi(gv).$$

It is now easy to show that T is a G -module homomorphism and $T^2 = T$. Note that $Tu = u \quad \forall u \in U$. Since T being a projection and $W = \text{Ker } T$ being G -module we get the required decomposition. $\square$

Finally, we obtain the proof of Maschke's theorem by the same argument as that of Lemma 2.

2 Building Representations out of existing ones

Before proceeding further, we would like to explore the representations of direct sums, tensor products and homomorphism spaces.

Let V and W be 2 representations of a finite group G , then the dual representation $\rho^* : G \rightarrow \text{GL}(V^*)$ is defined by

$$\rho^*(g)f(v) = f(\rho_{g^{-1}}(v)).$$

Let us have few insights behind this definition. We know that if ϕ is a linear operator on V , then the dual operator $\phi^* : V^* \rightarrow V^*$ is defined by $\phi^*(f)(v) = f(\phi(v))$ and if w.r.to some basis ϕ has a matrix A , then ϕ^* has matrix A^t w.r.to the dual basis. Now notice that dual map is a contravariant functor, i.e., $\phi^* \circ \psi^* = (\psi \circ \phi)^*$. But in terms of dual representation what we need is the covariant functor if you think $\phi = \rho_g$ and $\psi = \rho_h$ for some $g, h \in G$. Therefore consider the map $G \rightarrow G^{op} \rightarrow \text{GL}(V^*)$ defined by $g \rightarrow g^{-1} \rightarrow \phi_{g^{-1}}^*$. This is the required representation.

Also $V \otimes W$ is a representation via

$$g(v \otimes w) = gv \otimes gw.$$

Finally we conclude this section by describing the representation for $\text{Hom}(V, W)$. For this we identify $V^* \otimes W = \text{Hom}(V, W)$ by the map $(f, w) \rightarrow f_w$ with $f_w(v) = f(v) \cdot w$. Under this identification, one could verify that for $\phi \in \text{Hom}(V, W)$, $(g\phi)(v) = g\phi(g^{-1}v)$. Therefore the dual representation is, in turn, a special case of this; when $W = \mathbb{C}$.

Unraveling the representation $\text{Hom}(V, W)$: Let

$$\psi : V^* \otimes W \rightarrow \text{Hom}(V, W)$$

be the isomorphism

$$\psi(f \otimes w) = f_w \quad \text{such that} \quad f_w(v) = f(v) \cdot w.$$

Let $\phi \in \text{Hom}(V, W)$ with $\psi(f \otimes w) = \phi$ for some $f \in V^*$, $w \in W$. We want to find $(g \cdot \phi)(v)$. Note that

$$\psi(g \cdot (f \otimes w)) = g \cdot \psi(f \otimes w) = g \cdot \phi.$$

Now,

$$\begin{aligned} (g \cdot \phi)(v) &= (\psi(g \cdot (f \otimes w)))(v) \\ &= (\psi((g \cdot f) \otimes (gw)))(v) \\ &= (g \cdot f)(v)gw \\ &= f(g^{-1}v)gw \\ &= g \cdot (f(g^{-1}v)w) \\ &= g\phi(g^{-1}v). \end{aligned}$$

3 Schur's Lemma

Theorem 3. *If V and W are irreducible representations of G and $\phi : V \rightarrow W$ is a G -module homomorphism, then*

1. *Either ϕ is an isomorphism, or $\phi = 0$.*
2. *If $V = W$, then ϕ is a scalar (scalar from $\mathbb{C}$) multiple of identity.*

The proof is very easy so that we skip that. As an immediate consequence of this theorem, we see that all the irreducible representations of an abelian group is one dimensional. This is because if G is abelian and V is an irreducible representation, then the map $\rho_g : V \rightarrow V$ is a G -module homomorphism and by Schur's lemma every element $g \in G$ acts on V by a scalar multiple of the identity. Thus every subspace of V is invariant; so that V must be one dimensional.

4 Representations of the Symmetric group S_3

Here we will classify all the irreducible representations of S_3 without using any character theory. We can verify that the following representations of S_3 are irreducible.

1. The *trivial representation*.
2. The *alternating representation* defined by setting $gv = \text{sgn}(g)v$ for $g \in S_3, v \in \mathbb{C}$.
3. Now we look at 2 dimensional irreducible representations. Before that consider the natural permutation representation in which S_3 acts on $\mathbb{C}^3$ by permuting the coordinates, i.e., $ge_i = e_{g(i)}$. But this representation is not irreducible because it has a one-dimensional sub-representation namely the span of $(1, 1, 1)$. Now we take the orthogonal complement of this subspace which is also a representation of S_3 . What is that? It is obviously the subspace

$$V = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : z_1 + z_2 + z_3 = 0\}.$$

This is also an irreducible representation other wise it will have a one dimensional invariant subspace which will correspond to an eigen vector for all the elements of S_3 . But the only eigenvectors of $\mathbb{C}^3$ for all $g \in S_3$ are the scalar multiples of $(1, 1, 1)$. Thus this representation is irreducible and this representation is called the *standard representation* of S_3 .

We claim that these three are all the irreducible representations of S_3 . We do this by analyzing the action of the cyclic subgroup of S_3 on an arbitrary representation W of S_3 . Let σ be a generator (3 cycle) of

the cyclic subgroup of S_3 . Then since σ has order 3, the linear operator ϕ_σ of S_3 on W is diagonalizable and so that W is spanned by the eigenvectors v_i for the action of σ whose eigenvalues are of course all powers of a cube root of unity $\omega = e^{2\pi i/3}$.

Next we ask how the remaining elements of S_3 act on W . We know that if τ is a transposition, then σ and τ generate S_3 with the relation $\tau\sigma\tau = \sigma^2$. Now note that

$$\sigma(\tau(v)) = \tau(\sigma^2(v)) = \tau(\omega^{2i} \cdot v) = \omega^{2i} \cdot \tau(v)$$

where v is the eigenvector of σ with the eigenvalue ω^i . In conclusion, if v is an eigenvector for σ with eigenvalue ω^i , then $\tau(v)$ is again an eigenvector for σ , with eigenvalue ω^{2i} .

Now we come to the end of the last stage of classifying the irreducible representations of S_3 . Let σ be as before with eigenvector v and eigenvalue ω^i . If $\omega^i \neq 1$, then $\omega^{2i} \neq \omega^i$ and $\tau(v)$ is an eigenvector of σ with eigenvalue ω^{2i} . Then v and $\tau(v)$ are linearly independent and they span a 2 dimensional S_3 -invariant subspace V' of W which is isomorphic to the standard representation.

If the eigenvalue of σ with eigenvector v is 1, then $\tau(v)$ may or may not be independent of v . If it is not, $\tau(v) \pm v$ and v spans a one-dimensional subrepresentation of W isomorphic to the trivial representation if $\tau(v) = v$ and isomorphic to the alternating representation if $\tau(v) = -v$. If $\tau(v)$ and v are independent, then $v + \tau(v)$ and $v - \tau(v)$ span one-dimensional subrepresentations of W isomorphic to the trivial and alternating representations, respectively. It must be clear at this point that any irreducible representation of S_3 must be one of the three forms defined above.

5 Character Theory

Character: V -representation of G , $\chi_V : G \rightarrow \mathbb{C}$ be the homomorphism

$$\chi_V(g) = \text{Tr}(g|_V)$$

χ_V is constant on conjugacy classes – therefore it is called a *class function*. Note that

$$\begin{aligned} \chi_{V \oplus W} &= \chi_V + \chi_W, & \chi_{V \otimes W} &= \chi_V \cdot \chi_W \\ \chi_{V^*} &= \overline{\chi_V}, & \chi_{\Lambda^2 V}(g) &= \frac{1}{2} [\chi_V(g)^2 - \chi_V(g^2)]. \end{aligned}$$

5.1 Characters of irreducible representations of S_3

Conjugacy classes: (1), (12), (123)

	(1)	(12)	(123)
i) Trivial U	1	1	1
ii) Alternating U'	1	-1	1
iii) Standard V	2	0	-1

Note that the permutation representation decomposes

$$\mathbb{C}^3 = U \oplus V$$

$$\begin{aligned} \chi_V((1), (12), (123)) &= \chi_{\mathbb{C}^3}((1), (12), (123)) - \chi_U((1), (12), (123)) \\ &= (3, 1, 0) - (1, 1, 1) = (2, 0, -1). \end{aligned}$$

Now If $W = U^{\oplus a} \oplus U'^{\oplus b} \oplus V^{\oplus c}$, then $\chi_W = a\chi_U + b\chi_{U'} + c\chi_V$.

Note: $\chi_U, \chi_{U'}, \chi_V$ are linearly independent (we'll prove this!). Thus W determined up to isomorphism by its character χ_W .

Q: What is the decomposition into irreducibles of $V \otimes V \cong ?$ where V as above.

$\Rightarrow$ Character of χ_V^2 :

	(1)	(12)	(123)
χ_V	2	0	-1
χ_V^2	4	0	1

Let $\chi_{V \otimes V} = a\chi_U + b\chi_{U'} + c\chi_V$:

$$a + b + 2c = 4, \quad a - b = 0 \Rightarrow a = b$$

$$a + b - c = 1, \quad a = b$$

$$2a + 4a - a = 6a = 6 \Rightarrow a = 1, \quad b = 1, \quad c = 1$$

Thus:

$$V \otimes V \cong V \oplus U \oplus U'.$$

Q: What is the decomposition of $V^{\otimes n}$?

$\Rightarrow$ Let $V^{\otimes n} \cong U^{\oplus a} \oplus U'^{\oplus b} \oplus V^{\oplus c}$. Then $\chi_V^n = (2n, 0, (-1)^n) \Rightarrow a + b + 2c = 2^n, \quad 2a - c = (-1)^n$ and $a = b$. Solving, we get

$$a = \frac{1}{3} (2^{n-1} + (-1)^n) = b, \quad c = \frac{1}{3} (2^n + (-1)^{n-1}).$$

5.2 Some Projection Formulas

Let V be any G -module (finite dimensional). Let

$$V^G = \{v \in V : gv = v \quad \forall g \in G\}.$$

The map $\phi : V \rightarrow V$ defined by

$$\phi = \frac{1}{|G|} \sum_{g \in G} g$$

is a G -module homomorphism.

Theorem 4. *The map ϕ is a projection of V onto V^G , i.e., $\text{Im } \phi = V^G$ and $\phi^2 = \phi$.*

Q: What is the number of trivial representations appearing in the decomposition of a G module V ?
 $\Rightarrow$ Clearly V^G is a G -module. Let $V = U^{\oplus m} \oplus W$, U = trivial representation, and then $V^G = U^{\oplus m}$.
Thus:

$$m = \dim V^G = \text{tr}(\phi) = \frac{1}{|G|} \sum_{g \in G} \text{tr}(g) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g). \quad (1)$$

Thus:

$$\sum_{g \in G} \chi_V(g) = 0 \quad \text{for an irreducible } V \text{ (other than trivial).}$$

This is because if V is irreducible, $V^G = V$ or 0 . If $V^G = V$ then trivial. If $V^G = 0$, then the statement follows because then dimension is zero.

Theorem 5. *If V is irreducible, $\dim \text{Hom}(V, W)$ is the multiplicity of V in W (here $\text{Hom}(V, W)$ is the space of G -module homomorphisms from V to W).*

Proof. Let $W \cong V^{\oplus a} \oplus (\oplus V_i^{m_i})$ where V_i are irreducible representations other than V , occurring in the decomposition of W . Then

$$\text{Hom}(V, W) = \text{Hom}(V, V^{\oplus a} \oplus W') = \text{Hom}(V, V)^{\oplus a} \oplus \text{Hom}(V, W')$$

where $W' = \oplus V_i^{m_i}$. The statement follows from Schur's lemma. $\square$

Similarly, if W is irreducible, $\dim \text{Hom}_G(V, W)$ is the multiplicity of W in V . If both V and W are irreducible:

$$\dim \text{Hom}_G(V, W) = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{if } V \not\cong W. \end{cases} \quad (2)$$

We know that $\text{Hom}(V, W) \cong V^* \otimes W$. Thus

$$\chi_{\text{Hom}(V, W)}(g) = \overline{\chi_V(g)} \chi_W(g).$$

Thus, for two irreducible G -representations V and W , if $V \not\cong W$, then by (1) $\text{Hom}_G(V, W) = 0$. Hence $\overline{\chi_V} \chi_W = 0$. If $V \cong W$, then

$$\dim \text{Hom}_G(V, W) = 1,$$

Then from (1) it follows that

$$1 = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g).$$

[Note that to follow from (1), we have the fact that

$$\text{Hom}(V, W)^G := \{\phi \in \text{Hom}(V, W) : g \cdot \phi = \phi \ \forall g \in G\} = \text{Hom}_G(V, W).$$

For a proof observe that if $\phi \in \text{Hom}(V, W)^G$, $\phi(g \cdot v) = (g \cdot \phi)(gv) = g \cdot \phi(g^{-1}gv) = g \cdot \phi(v)$ and if $\phi \in \text{Hom}_G(V, W)$, then $(g \cdot \phi)(v) = g\phi(g^{-1}v) = gg^{-1}\phi(v) = \phi(v)$ (recall how g acts on $\text{Hom}(V, W)$ from section 2).] Thus if V and W are irreducible G -modules, then

$$\frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{if } V \not\cong W. \end{cases} \quad (3)$$

Let $\text{Class}(G) = \text{space of class functions on } G$. For $\alpha, \beta \in \text{Class}(G)$, define the inner product:

$$\langle \alpha, \beta \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\alpha(g)} \beta(g).$$

Then (3) says that the characters of irreducible representations of G are orthonormal.

Corollary 1. *The number of irreducible representations of G is at most the number of conjugacy classes.*

Proof. The characters of irreducible representations are orthogonal $\Rightarrow$ linearly independent. Also $\dim \text{Class}(G) = \#\{\text{conjugacy classes of } G\}$ (basis: δ_C , the function which is 1 on the conjugacy class C and 0 elsewhere). $\square$

Corollary 2. *If V, W are representations of G , then $V \cong W \iff \chi_V = \chi_W$.*

Proof. " $\Rightarrow$ " is clear. " $\Leftarrow$ " follows by decomposition and linear independence of characters. $\square$

Corollary 3. *A representation V is irreducible iff $\langle \chi_V, \chi_V \rangle = 1$.*

Proof. If V is irreducible:

$$\langle \chi_V, \chi_V \rangle = \frac{1}{|G|} \sum_{g \in G} |\chi_V(g)|^2 = 1$$

by (3). Converse follows from the next corollary. $\square$

Corollary 4. *If $V \cong \bigoplus_i V_i^{\oplus a_i}$ where V_i are irreducible, then*

$$\langle \chi_V, \chi_V \rangle = \sum_i a_i^2,$$

and $a_i = \langle \chi_V, \chi_{V_i} \rangle$.

Regular representation R on $\mathbb{C}[G]$: $g \cdot h = gh$ (left multiplication). If $g = e$, $\chi_R(g) = |G|$. If $g \neq e$, $\chi_R(g) = 0$.

If $G \neq \{e\}$, R is not irreducible. This is because if R is irreducible, $\langle \chi_R, \chi_R \rangle = 1$, i.e.,

$$\frac{1}{|G|} (|G|^2) = |G| = 1 \Rightarrow |G| = 1 \Rightarrow G = \{e\}.$$

Let $R = \bigoplus V_i^{a_i}$, where V_i are distinct irreducibles. Then

$$a_i = \langle \chi_R, \chi_{V_i} \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_R(g)} \chi_{V_i}(g).$$

Since $\chi_R(e) = |G|$, $\chi_R(g) = 0$ for $g \neq e$,

$$a_i = \frac{1}{|G|} \cdot |G| \cdot \dim V_i = \dim V_i.$$

Corollary 5. *Any irreducible representation V of G appears in the regular representation $\dim V$ times.*

Corollary 6. *We have*

$$|G| = \dim(R) = \sum_i (\dim V_i)^2.$$

Proof. From $R = \bigoplus V_i^{(\dim V_i)}$,

$$|G| = \dim R = \sum_i \dim V_i \cdot \dim V_i = \sum_i (\dim V_i)^2. \quad \square$$

Note that if $g \neq e$, $\chi_R(g) = 0 \Rightarrow 0 = \sum_i \dim V_i \cdot \chi_{V_i}(g)$.